

16 Hydroxy compounds

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Alcohol oxidation depends on carbon bearing -OH: primary gives aldehyde then acid; secondary gives ketone; tertiary resists mild oxidation. Conditions matter: distil aldehyde to prevent further oxidation, reflux for acid. Phenol is more acidic than an alcohol because phenoxide ion is resonance stabilised.

Worked example: follow the reasoning

Oxidising propan-2-ol with acidified dichromate gives propanone; orange dichromate turns green.

Quick check · multiple choice

A Which alcohol typically oxidises to a ketone?

[1]

- A Primary
- B Tertiary
- C Phenol only
- D Secondary

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Outline conditions and product for oxidising ethanol to ethanoic acid rather than ethanal.

[3]

Answers and explanation

Quick check: D. Secondary

Question 1 [3 marks]

Use excess acidified dichromate(VI) [1]; heat under reflux [1]; product ethanoic acid [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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